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Students Lack Effective Collaboration Skills

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Students Lack Effective Collaboration Skills

By

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Master's Project

Submitted to the College of Education

At Grand Valley State University

In partial fulfillment of the

Degree of Master of Education

Instruction and Curriculum



The signatures of the individuals below indicate that they have read and approved the project of Robin VanderWel in partial fulfillment of the requirements for the degree of Master of Instruction and Curriculum – Secondary Education Program.

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Abstract

It has been seen through field experience and through research that students lack the ability to effectively learn through collaboration. The research has shown that there tends to be a lack of authentic learning when using collaboration projects/activities because students do not possess the skills that are needed to collaborate effectively. The ability to collaborate needs to be explicitly taught through modeling and scaffolding. This project examines research on the importance of collaborative learning and the required skills that students need to be successful. The project used research to develop a professional development session that allows teachers to see the importance of collaborative learning skills and provides resources that can be used in secondary education to explicitly teach the 21st century skill of collaboration to students. The professional development provides group norms, group roles, group contract, conflict resolution strategies, daily reflection for students, and a collaboration lesson. The session will be considered successful if teachers are able to use the resources in the classroom and provide qualitative data of students developing collaborative learning skills.

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Chapter 1 Introduction

Problem Statement

Many students lack effective collaboration skills. In order to effectively collaborate, students need to access higher-order thinking skills. These skills include critical thinking, creativity, problem solving, and communication and are crucial for effective collaboration within the classroom (Siew & Basari, 2024). Collaboration skills are needed for a classroom to transition from a teacher directed classroom to a collaborative learning classroom. According to Liebech-Lien and Sjølie (2021), “Collaboration is widely considered to be a valued educational practice in schools” (p.213).

Students may lack effective collaboration skills due to not seeing collaboration with peers as important and view teaching each other peers as a waste of time, and becoming frustrated that all group members are not fully participating (Siew & Basari, 2024). Students may also be unaware of how to effectively collaborate. Students have to develop a foundational knowledge of the content within classrooms, they also need to know how to work within a group and effectively collaborate in order to learn from each other and provide effective feedback. It should not be assumed that students automatically have collaboration skills; this needs to be modeled, scaffolded, and explicitly taught for students (Liebech-Lien and Sjølie, 2021).

Importance and Rationale of the Project

The reason that students need to develop effective collaboration skills because it is classified as a higher order thinking skill (HOTS) and a 21st century skill that students will be expected to have as they progress through their academic and professional

career (Lu et al., 2021; Bielik et al., 2022; and AIAli., 2024). In a study done by Bielik et al., they discussed the importance of having students engage in collaborative learning activities; “The importance of teaching science as an ongoing collaborative endeavor in which students build ‘21st-century skills’ by considering diverse perspectives and by learning to work together to solve problems” (p.425). Twenty first century skills, often referred to as soft skills, tend to have collaboration as one of the core skills required for students to develop; other skills can include creativity, critical thinking, and communication (Liebech-Lien and Sjølie, 2021).

In a study done by AIAli (2024), which took place in Saudi Arabia, they used a pedagogical approach of Problem and Project Based Learning (PoPBL).

“PoPBL is derived from the Problem-Based Learning (PBL) model and is guided by core principles, including student-centeredness to motivate and engage learners, a focus on the learning process of finding solutions, project-based activities with actionable goals, exemplarity, and the promotion of group work, teamwork, social skills, and communication abilities” (AIAli, 2024, p. 422).

From this study, AIAli supported the idea that student-inquiry pedagogy has the potential to improve communication skills. AIAli (2024) discovered that, “Individuals need to be able to express their ideas clearly, collaborate with others, and engage in constructive dialogue” (p. 422). Being able to communicate effectively is a needed 21st century skill that is highly marketable; this includes orally and written communication (AIAli, 2024). In order for a student to collaborate with a peer, they have to practice

communication with others that have the same and/or different perspectives. These skills are needed for students to develop in a safe place of school, so that they will be able to continue to use them within the workforce (AlAli, 2024).

In order for students to develop collaboration skills, the teacher needs to provide opportunities for the students to practice those skills; this can be done through collaborative inquiry-based learning (CIBL) or project based learning (PBL). In order for students to accurately participate in CIBL or PBL, they need to work with peers to discuss ideas and solutions (Bielik et al., 2022). When students are given the opportunity to work collaboratively with each other, they will be able to learn from each other; possible learning opportunities for students when they collaborate and actively engage with peers is how “to support, motivate, and enhance one another’s efforts to learn as much as possible” (Siew & Basari, 2024; p.287). Siew and Basari saw this in their study with students that have a variety of academic ability, “Students with lower achievement are helped by their other group members to create a complete picture with unique and interesting information and its meaning is understood by each member” (p.287). This suggests that collaborative work allows students to come together in a social environment that has potential for a better educational experience; that allows students to understand the content and actively discover the importance of communication(Siew & Basari, 2024).

When students are actively engaged in collaborative learning, there is potential that students can become more invested in their learning and ultimately become more motivated to understand the content (Najjar and Daher, 2023). Ways that teachers can help with motivation is to look at the student perspective of things; such as

understanding and responding to their perspective (Ryan & Deci, 2020). Ryan and Deci (2020) also discussed a way that teachers can enhance student motivation by providing a meaningful rationale behind the reason they are doing a project or learning a particular concept and allow students meaningful choices and tasks to engage the interests of the students. By examining the motivation and engagement of students, the students potentially would gain more knowledge of the content instead of just wanting to finish the work as soon as possible (Puntambekar et al., 2023).

Background of the Project

“The working together of two or more students with the same goal in mind can generate better achievement and result for classroom teaching” (Bada & Jita, p.2, 2022). This supports the notion that classrooms are shifting towards more student centered and inquiry based learning to allow for more authentic learning. The current situation tends to be a more teacher centered way of teaching, this is a more traditional aspect of schooling and potentially contributes to negative student feelings towards the classroom (Long et al., 2021). To move away from a more teacher centered direct instruction classroom practices and more towards a more student centered classroom that focuses on CIBL or PBL, teachers will need to examine classroom practices and procedures to allow for collaboration skills to be taught and practiced (Bielik et al., 2022). Teacher directed or student centered practices and pedagogy has potential to allow for student achievement, what determines if there is achievement tends to be the frequency of activities, but also the amount of guidance that is provided with the activities (Long et al., 2021)

The author of this project has been in the process of revamping curriculum used to allow for more collaborative learning to occur to better meet the standards of Next Generation Science Standards (NGSS). However, as developing projects and having students work through them; the author has seen that students need more guidance on how to effectively work together and not just 'state' answers, not have one person do majority of the work, and/or how to make sure all students are actively engaged within the project.

If a student is used to a traditional teacher-focused learning environment, the transition to engaging in collaborative learning activities can be challenging for the student(s); especially if the activities are inquiry based (Bielik et al., 2022). Siew and Basari (2024) discovered in their study that as a classroom is transitioning to a more PBL or CIBL style of teaching and students are unaware how to effectively collaborate with each other, which has potential for students to develop negative perspectives on collaboration; this can happen if not all members are participating or there is a lack of instruction/understanding on what exactly to do. Students can have developed negative feelings towards cooperative learning if the teacher is unaware of how to teach and use the pedagogy accurately (Bada and Jita, 2022). In order to avoid negative perspectives on collaborative learning, teachers will need to assess student collaboration skills; similar to the way that teachers assess content knowledge; through formal and informal observations.

When students are working with CIBL or PBL curriculum, they need to be able to exchange ideas about the problem, solution, and any decisions that are needed to proceed with the content (Puntambekar et al, 2023). In order for students to be

successful with these types of collaboration and collaborative learning there needs to be communication processes and interactions between peers. Students need to practice communication skills at different levels based on their expertise of collaborative learning. This was supported in the study by Puntambeker et al., “Explaining one’s ideas to others may result in greater cognitive engagement and the building of more connections between ideas that can positively influence learning” (p.560). A skill that was recommended from the study done by Puntambeker et al., was the importance of the groups creating norms of collaboration that can be adaptive and changes based on the students’ ability and knowledge of how to have an effective collaborative discussion.

Students need to know how to work within a group and effectively collaborate in order to learn from each other and provide effective feedback. It should not be assumed that students automatically know how to do this. It needs to be modeled and scaffolded for the students (Liebech-Lien & Sjølie, 2021). The lack of knowing how to work effectively with peers to use inquiry and to develop a model will hinder their ability to understand the concept through their collaborative student led model developed by the teacher. However, the more time that is spent working with and learning how to cooperate/collaborate with each other, students will learn the essential skills that are needed for collaboration to be effective (Bada & Jita, 2022).

Statement of Purpose

The purpose of this project is to work on students’ collaboration skills through modeling, scaffolding, and explicit instruction in order for students to develop the needed 21st century skill.

Objectives of the Project

Liebech-Lien and Sjolie (2021) stated that, “the renewed focus on collaboration skills represents a shift away from collaboration as a way of organizing and teaching a subject, and towards the notion of collaboration as an educational outcome that needs to be learned explicitly” (p.214). This helps explain the importance that students need to be taught how to work collaboratively with their peers. The objective of this project is to: develop collaboration discussion skills, group assigned roles, and develop collaboration conflict strategies. This project will equip teachers with the knowledge on how to model and scaffold collaborative learning skills. “This theme of collaboration skills being taken for granted also includes statements describing how teachers have limited experience working with collaboration skills from their own education” (Liebech-Lien & Sjolie, 2021, p.22).

When students are collaboratively learning and sharing perspectives, there creates a potential for conflict to arise. The author of this project has seen conflict arise within groups and did not have a structure in place that would allow students to solve the problems without the teacher's involvement. According to a study by Liebech-Lien and Sjolie (2021), some students struggle to collaborate with their peers due to lack of motivation, lack of academic content, struggle making relationships, and overall students that do not work well together.. Within the study, many teachers stated that they were not sure on how to guide students that struggle with collaboration conflict.

Definition of Terms

- **21st century skill:** These are skills that are needed in today’s modern workforce for a student to be successful. These skills tend to go above the content

standards and include critical thinking, problem solving, creativity, collaboration, communication, and technology skills.

- **Collaborative inquiry-based learning (CIBL):** Collaborative Inquiry Based Learning is a curriculum based approach when two or more students learn together by engaging in inquiry based learning, often includes given a real life situation where they need discussion to develop a working solution.
- **Collaboration skills:** A 21st century soft skill that is needed within the workforce. A skill that allows one to effectively and respectfully work with others and take shared responsibility of the work (Liebech-Lien and Sjølie, 2021).
- **Higher Order Thinking Skills (HOTS):** Skills that are above memorization of facts. Examples of HOTS include critical thinking, problem solving, making inferences, creating new solutions, and collaboration.
- **Inquiry Based:** Students are given the foundational knowledge to make connections between the concepts and principles in order to apply their knowledge (Najjar and Daher, 2023).
- **Problem Based Learning (PBL):** Focuses on solving real world problems and is based on student-centered and inquiry driven design principles (Bielik et al., 2022).
- **Scaffolding:** A teacher gradually removing guidance and support as students learn and become more competent with the content.
- **Student Centered:** Students working through the concepts and developing solutions and modeling to demonstrate their learning and to show growth with guidance from the teacher (Long et al., 2021).

- **Teacher Centered:** When there is mostly direct instruction with the teacher lecturing and doing majority of the talking while the role of the student is to take notes to use at a later time to memorize the content (Long et al., 2021).

Scope of Project

This project will not be addressing intentional versus unintentional grouping, such as group balancing based on gender, academic ability, etc. The age of students is intended for 6th through 8th grade; however has potential to be adapted for lower or higher grades. Factors that are out of the researchers control are any time constraints that may arise, educator 'buy-in' in the importance of collaborative learning skills, and students' attitudes towards collaborative learning with peers. The overall project will create a professional development that allows for teachers to be better equipped to develop a collaborative learning environment by focusing on the skill of collaborative discussion, creation of group roles and norms, and how students can settle conflict without teacher involvement.

Chapter 2: Literature Review

Introduction

Collaboration allows students to develop skills that allow for life long learning and the ability to see world wide perspectives (Gratton, 2019). The problem is that students lack collaboration skills. According to Najjar and Daher (2023), "The primary role of teaching science is not to teach all the facts, it is to provide students with sufficient basic knowledge to enable them to discover knowledge on their own" (p.1175). This supports the pedagogy of collaborative learning. However, when introducing collaboration students need to know the procedures of how to collaborate (Liebech-Lien & Cunha de

Araújo, 2020; Naggar and Daher, 2023). The main components of collaborative learning that needs to be addressed in order to have students develop collaboration skills are: preparing the classroom layout, the grouping of students, preparing and enhancing student skills of interaction with each other, the teacher understanding the shift in their role, and the preparation of the project/activity (Blatchford et al., 2003). This paper will focus on the preparation of student collaboration skills and the shift in the teacher role.

To move away from a more teacher centered direct instruction classroom practices and more towards a more student centered classroom that focuses on CL or PBL, teachers will need to examine classroom practices and procedures to allow for collaboration skills to be taught and practiced (Bielik et al., 2022). Students need to consistently participate within collaborative learning in order to improve their collaboration skills and students need guidance to enhance their collaboration skills as well (Lu et al., 2021).

This chapter will explain how collaborative learning is linked to social constructivism and constructive theory. It will then use qualitative and quantitative research to look at the aspects of collaboration skills by examining collaborative learning and project based learning that impacts 21st century skills, the benefits of collaboration, collaboration drawbacks, collaborative classrooms, and conflict that arises. These aspects will be addressed to examine how to create an environment that allows for the development of collaboration skills.

Theory/Rationale

Collaborative learning is when there are two or more students that are working together to solve a problem in order to develop a better understanding of the content

(Bada & Jita, 2022). “Collaborative skills are defined as a partnership between two or more students” (Bevo et al., 2025, p. 86). This is inline with social constructivism and constructive theory. Constructivist theory requires active learning and students learn more by doing (Bell, 2010; Naidoo & Mabaso, 2023) Social constructivism is the idea that knowledge and the importance of the content is socially created with effort and interaction from other people (Naidoo & Mabaso, 2023; Lange et. al., 2016). Social constructivists tend to view students as contributing to the learning through socialization (Buchs et al., 2016). Within a social environment, it can develop a student’s learning and motivation (Blatchford et al., 2003). Social constructivism needs social interaction; to be a well-developed social constructivist teacher, there needs to be a shift in the type of learning that is occurring within the classroom (Naidoo & Mabaso, 2023). This shift should be from teacher-directed learning towards more student centered learning; student centered learning allows students to be actively involved within the content and collaborative skills allows students to be active learners.

Collaboration and social constructivism theory have been supported by multiple studies by seeing that students tend to perform better when working in groups, able to learn and retain knowledge better than working individually. (Liebech-Lien & Cunha de Araújo, 2020; Gillies & Boyle, 2010; Lange et. al., 2016; Bevo et al., 2025). A study done by Care et. al. (2016), supported this idea as well, “Social interactions make explicit a student’s understanding and it has been found that students process information differently when they work in groups compared to working independently” (p.252). When a curriculum is shifted towards more student centered, it is important that the students are learning the content; but the teacher should also be concerned with

how well a student is able to work with others (Naidoo & Mabaso, 2023). Collaborative learning provides students opportunities to learn from each other through supporting, motivating, and enhancing each others' learning experiences which supports the idea of constructivism and social constructivism (Siew & Basari, 2024).

Research/Evaluation:

Collaborative Learning and Project Based Learning Curriculum:

Project based learning (PBL) and collaborative learning (CL) is recommended for students to practice 21st century skills (Roshid and Haider, 2024). Collaborative learning is when there are two or more students that are working together to solve a problem in order to develop a better understanding of the content and is considered an active learning curriculum and allows for a more student centered classroom (Bada & Jita, 2022; Liebech-Lien & Cunha de Araújo, 2020). Project based learning is more student centered and "PBL promotes social learning as children practice and become proficient with the twenty-first-century skills of communication, negotiation, and collaboration" (Bell, 2010, p. 40). Students' development of collaborative learning skills increase with an increase of PBL and CL (Siew & Basari, 2024).

The curriculum tends to focus on 'real life' problems/projects (Siew and Basari, 2024). "Real-world projects deepen learning for students" (Bell, 2010, p.42). By bringing in 'real-life' aspects into the curriculum tends to alter the role of the teacher and of the student; students will become more active in their learning (Najjar and Daher, 2023). Making the content related to 'real world issues', it potentially has the ability for students to understand the why behind the content. If students understand the content and how something occurs, but lack why it is important or the impact it has on society; then the

student ultimately has not grasped the knowledge of the content (Gratton, 2019). The use of PBL focuses on finding solutions and performance instead of just recall (Bielik et al., 2022).

When students are working with collaborative learning or project based learning curriculum, they need to be able to exchange ideas about the problem, solution, and any decisions that are needed to proceed with the content (Puntambekar et al, 2023). As students are collaborating with each other as Puntambekar et al. stated, the students start to learn from each other and there becomes an understanding between the students (Bada & Jita, 2022).

“In comparison to simple tasks, the likelihood that one individual alone can complete complex tasks is diminished, and consequently the need for individuals who can bring together the required sets of skills and resources comes about” (Care et. al.,) “Working together may require a different set of skills than does working alone, both in terms of interpersonal dynamics as well as in cognitive processes” (Care et. al., p.251).

When students are engaged and active with their learning; there tends to be a higher retention rate and allows for more authentic learning; however students need skills to make this achievable (Bielik et al., 2022).

Collaborative learning and project based learning is used to increase a student's ability to collaborate and learn content through collaboration (Liebech-Lien & Cunha de Araújo, 2020; Bielik et al., 2022). When teachers are shifting to a more of a collaborative or a PBL curriculum, they need to learn new skills as well; such as how to

be more of a facilitator, how to design authentic collaborative learning activities, and how to teach collaboration skills (Liebech-Lien & Cunha de Araújo, 2020). The teacher needs to develop more of a facilitator role and will need to know when to step in (Najjare and Daher, 2023; Bell, 2010; Blatchford et al., 2003; Gratton, 2019; Bada & Jita, 2022; Siew and Basari, 2024). If the collaborative learning activity has not had the accurate amount of guidance there is the potential for students to fail to learn anything and can introduce misconceptions about the content (Long et al., 2021). Knowing the amount of guidance that is required requires scaffolding collaboration instructions. Students can develop negative feelings towards cooperative learning if the teacher is unaware of how to teach and use the pedagogy accurately (Bada and Jita, 2022).

Benefits of Collaborative Work Within and Outside of the Classroom

“In the future, children must enter a workforce in which they will be judged on their performance. They will be evaluated not only on their outcomes, but also on their collaborative, negotiating, planning, and organizing skills” (Bell, 2010, p. 43). The skills mentioned by Bell (2010) are considered a higher order thinking skill (HOTS) and is a 21st century skill (Lu et al., 2021). Collaborative learning tends to be a mix of a variety of 21st century skills; these skills can be classified as critical thinking, problem solving, decision making, and collaboration (Care et.al, 2016). “Decision-making and collaboration are the essential skills for students to effectively navigate the challenges of the 21st century” (Bevo et al., 2025, p. 86).

When students are collaborating together, students will bring their strengths to the group to help the overall success of the project; similar to how in the ‘real world’ people use their strengths/talents within their job (Bell 2010). “Collaboration allows

students to learn how to communicate, cooperate, and coordinate efforts toward a common objective effectively” (Bevo et al., 2025, p. 92). When they are collaborating together, there is a need for communication. Communication is considered a soft skill. “‘Soft skills’ is an umbrella term covering various survival skills such as communication and interpersonal skills, emotional intelligence, leadership qualities, team skills, negotiation skills, time and stress management and business etiquettes” (Deepa & Seth, p.7, 2013). A study done by Deepa and Seth (2013), found that hiring managers are not impressed with the workforce coming out of education and need to have better soft skills, as well as the background knowledge to learn the hard skills that are required by the specific job. Along with communication, collaboration allows students to practice their goal setting and time management skills that are needed for success in the workforce (Bell, 2010).

When students are learning, there are intrinsic and external motivational factors at play. Collaborative work has the ability to allow students to become more interstitially motivated about their learning (Gratton, 2019). When a student becomes intrinsically motivated, they tend to perceive themselves as participating in a task that allows for challenge, curiosity, and motivation (Lu et al., 2021). Evaluation from groupmates impacts a student’s extrinsic motivation during collaborative learning and can hold the student accountable for their participation (Lu et al., 2021). “Accountability to peers often has greater consequences and provides more motivation for students than if they were only responsible to the teacher” (Bell, 2010, p. 40).

“Students learn accountability with PBL through the daily goal setting, as well as through expectations of their peers” (Bell, 2010, p. 40). Goal setting allows students to

be able to manage their own time as a group. As students are learning how to collaborate, it should not be assumed that they know how to manage their time; the teacher should work with students regularly to make sure they are managing their time (Bell, 2010). When students are working within a group; their groupmates tend to hold each other accountable (Bell, 2010; Mertz et al., 2023). Students can hold each other accountable through self and peer reflection of a rubric that is associated with the project, teacher feedback observations, and evaluations done by peers (Bada & Jita, 2022; Bell, 2010; Mertz et al., 2023).

Students need to develop skills on how to organize their work and time and how to actively engage within the work (Blatchford et al., 2003). As students become more proficient with collaboration work, their time management and problem solving will increase as well (Care et.al, 2016). As students become more experienced with PBL and collaborative work they are able to self monitor and determine if they meant their goal for the day (Bell, 2010).

From multiple sources of research, it has been seen that collaborative learning and project based learning work allows for students to ask deeper questions, allow for peer mentoring, and opportunity for peers to provide effective feedback, which ultimately allows for a more authentic learning experience and remembrance of the content (Care et. al, 2016; Naggar and Daher, 2023; Bell, 2010; Roshid and Haider, 2024). As students are working with the content and exploring the problem at hand, it was found that students became more fascinated and more curious about the content; which allowed the teacher to increase the amount of content that was covered (Naggar and Daher, 2023).

In a study done by Gillies and Boyle (2010), they found that teachers that integrated collaborative learning within their curriculum had positive interactions; these positive interactions included students building relationships with peers, accepting their assigned group roles, learning how to take risks with the content and learning how to manage time. Gillies and Boyle (2010) study also showed the work that was generated by a majority of the groups was of pretty high quality.

Collaboration Drawbacks

Many teachers are aware of the benefits of collaborative learning and having students through projects that are related to 'real world' problems to learn the content (Bell, 2010). "Regarding the practice of collaboration and communication skills, it was found that teachers rarely provided opportunities for students to engage in pair or group work or present their work in groups" (Roshid and Haider, 2024, p.9). Many negative perspectives from teachers in regards to collaborative learning tends to stem from the lack of understanding how to use collaborative learning effectively within the classroom (Bada & Jita, 2022).

Many qualitative and quantitative studies have found reasons why teachers tend to move towards a teacher centered and direct instruction pedagogy. These reasons tend to include that collaborative work can be time consuming, make the room loud, increase in the amount of socialization, classroom seating is not ideal for group work, large class numbers, time constraints, and too many essential standards that need to be met within a school year (Roshid and Haider, 2024; Liebech-Lien and Cunha de Araujo, 2020; Liebech-Lien and Sjolie, 2021; Gillies and Boyle, 2010; Bielik et al., 2022; Siew & Basari, 2024). Other reasons why teachers are hesitant to have students collaborate is

because the teacher might lack the knowledge on how to create a collaborative culture, make sure all students participate, and all benefit from collaboration (Liebech-Lien & Cunha de Araújo, 2020; Gillies & Boyle, 2010; Bielik et al., 2022). According to a study done by Buchs et al. (2016), difficulties that elementary teachers have when implementing collaborative learning are: preparing and organizing students for the task, and having enough time to create projects/activities that align with the standards. This includes the creation of the classroom climate, the students' collaboration skills, and how the students' process the content (Buchs et al., 2016).

Teachers also expressed concerns with collaborative work in regards that students lack the foundational content knowledge that is needed, they lack the resources needed to teach PBL or CL, and the amount of time it takes to create effective authentic collaborative learning activities (Naidoo & Mabaso, 2023; Bada & Jita, 202). The development of collaboration skills takes a lot of time and practice for the student and the teacher (Bevo et al., 2025). Another concern that teachers expressed is the students adjusting to collaborative learning instead of individual work and it can be challenging for the student to move past only waiting for the answer (Gillies & Boyle, 2010; Bielik et al., 2022).

Collaborative Classrooms

“Teachers can enhance collaboration skills by creating an environment that promotes open communication, providing opportunities for students to work together, and offering guidance on effective strategies” (Bevo et al., 2025, p. 87). In order for a classroom to foster a culture of collaboration, students need to be taught good routines that allow for collaboration to occur (Liebech-Lien & Cunha de Araújo, 2020). A

teacher's way of thinking about the classroom from teacher centered to student centered is altered when completing authentic collaborative learning; this includes the creation of new procedures and expectations of students (Naggar and Daher, 2023). Collaboration between peers can only happen in a safe and respectful classroom; meaning that it is free from fear of failure, threats, and allows students to express their opinions freely (Bada & Jita, 2022; Morcom, 2022). The classroom environment needs to be established to allow students to feel comfortable working with peers that have a variety of perspectives (Lange et al., 2016).

The classroom set up is important to help create the environment of collaboration. A study done by Liebech-Lien & Cunha de Araújo (2020), found that if the classroom was organized in base groups of four it would allow for formal and informal collaboration. Liebech-Lien & Cunha de Araújo (2020) study discovered that having students sit in groups does not automatically transfer to students knowing how to collaborate and that teachers need "to focus on teaching them [students] how to collaborate and not keep changing the groups" (p.11). "When sitting together in an environment that does not support productive group work, pupils can be drawn off-task by social talk" (Blatchford et al., 2003,p. 156). This is supported by Hu and Chen (2024), that students rarely have productive talk when they are just sitting next to each other without any guidance on how to complete the collaborative work. There are times when teachers assume that when students are placed in groups, the students automatically know how to collaborate efficiently and resolve conflict with each other (Liebech-Lien and Sjølie, 2021). Teachers and students tend to lack the training needed to effectively participate in a classroom environment that is centered around collaboration (Blatchford

et al., 2003). Teachers can easily struggle with how much guidance to give groups during collaborative learning (Liebech-Lien and Sjølie, 2021). In order to allow collaborative learning to fully work, teachers will need to lower the risk of failure and create an environment that allows for mistakes, make the project/activity engaging, and know how much of the group work to scaffold (Blatchford et al., 2003).

The creation of classroom norms and group norms will help create a respectful environment (Morcom, 2022). When students are working collaboratively, the groups should create norms of collaboration that can be adaptive and change based on their ability and knowledge of how to have an effective collaborative discussion (Puntambekar et al, 2023). If students have experience with collaborative learning, there will be less scaffolding needed to help students create the group norms. According to Morcom (2022), a norms should have the following:

- “1. Participation and right to pass- given a choice to participate (psychological safety)*
- 2. Attentive listening- appreciate others’ perspectives*
- 3. Mutual respect-set high expectations for behavior*
- 4. Using lift ups/No put downs- promote a caring classroom*
- 5. Personal best/growth mindset- promote a positive mindset to achieve personal best” (p.1534).*

Along with the creation of norms, a study done by Mertz et al. (2023), recommends a creation of common collaboration rubric that is used to addresses the quality of content work, commitment to the group, level of leadership, communication, and amount of individual work that was done to help the success of the group. The rubric should be scaffolded to fit the needs of the students; the rubric is used for a student to evaluate themselves and their peers (Mertz et al., 2023). The rubric can be

used as a teaching tool on what good collaboration looks and sounds like (Mertz et al., 2023).

A way to allow students to feel comfortable in participating within collaborative learning is the scaffolding (Lange et. al., 2016). Having instruction scaffolded allows for cognitive growth and allows for students to bridge gaps between the knowledge and the skill (Bell, 2010). As Bell (2010) mentioned in their research, “scaffolds are temporary, and as the learner gains fluency in the skill, the supports are removed, leading to a self-confident and capable learner” (p.41). Students need to be taught how to collaborate with each other and this can be done by scaffolding the skills of collaboration (Bell, 2010; Bielik et al., 2022; Gillies and Boyle, 2010; Lange et. al., 2016; Blatchford et al., 2003). It should not be assumed that students automatically have collaboration skills; this needs to be modeled, scaffolded, and explicitly taught for students (Liebech-Lien and Sjølie, 2021).

When creating collaborative learning opportunities within the classroom, teachers need to attempt to motivate students to communicate and collaborate with the content by relating it to their daily lives in order to activate their interests (Lu et al., 2021). A collaborative classroom environment should allow opportunities for students to work on complex real life projects/problems that require teamwork, communication, and problem solving; all while focusing on the content standards required by the course (Bevo et al., 2025). Within an authentic collaborative learning project, students should have an individual aspect of the project and contribute with the group to create the bigger picture (Bada & Jita, 2022).

“Great collaborative learning tasks engage and excite students, require students to apply their knowledge, and demand the valuable participation of every group member” (Chapman & Roberts, 2015, p. 28). To help with participation, having students think about the problem/question individually first allows the student to prepare their contributions to the group and ultimately will increase the amount of participation (Lange et. al., 2016). In a study done by Siew and Basari (2024), students that do not participate in PBL pedagogy claimed to be bored. The cause of this could be attributed to the environment not requiring active learning and/or there was no clear guidance on the activity or how to collaborate with each other (Siew & Basari, 2024).

If there is a structure in place, collaborative learning allows all groupmates an opportunity to participate and learn (Buchs et al., 2016). This was done in a study by Liebech-Lien and Cunha de Araujo (2020), each student was given roles and responsibilities to learn particular content; this allowed for students to ensure individual accountability and interdependence. Giving students roles within the group was also seen in a study done by Mertz et al. (2023). Assigning students roles or having students create roles within collaborative learning “create opportunities and norms for collaboration within the groups” (Chapman & Roberts, 2015, p.29). According to Chapman & Roberts (2015), some potential roles that a collaborative group can have are principal investigator (makes sure sure member is fulfilling their role), supply manager (makes sure all the supplies that are needed are chosen and be using properly), data manager (records data), time manager (makes sure the group is on task and knows the amount of time left on the project), and strategic questioner (poses questions to make sure all possibilities are addressed) (Chapman & Roberts, 2015).

When students are working within a group during collaborative learning, they need to develop positive interdependence and individual accountability (Adl-Amini et al., 2024; Bevo et al., 2025). Positive interdependence is when the group realizes that they can not succeed without each other and individual accountability is knowing that you need to complete your role to help the group (Adl-Amini et al., 2024). In order for students to be effective with positive interdependence, there needs to be productive talk that allows students to explain their ideas and engage with other peer ideas; allowing them to deepen their understanding of the content (Hu & Chen, 2024). The development of productive peer talk should be scaffolded, practiced, and there should be self and peer reflection (Hu & Chen, 2024).

Conflict Arising

Being able to work in teams, while completing a common goal and resolving conflict is an essential to collaborative learning (Deepa and Seth, 2013). “Students need to employ their negotiation skills when conflicts arise in their efforts to reach the common goal” (Care et. al., 2016, p.253). Teachers need to equip students on how to manage conflict without teacher involvement and to supply strategies that allow students to consider the perspectives of others (Gillies & Boyle, 2010). For collaboration to occur, it is essential to have appropriate responses to others and recognize others (Care et. al, 2016). Students that are learning how to collaborate with peers will find it difficult to alter their behavior to their peer’s needs (Care et.al, 2016).

A type of conflict that arose for teachers within a study done by Gillies & Boyle (2010) was how students deal with a situation of when someone is not contributing and is just relying on the rest of the group or if someone is overtaking the group. This type of

conflict was also seen in multiple studies as well (Bell, 2010; Lange et. al., 2016; Siew & Basari, 2024). A reason why conflict might arise and/or certain groupmates might not participate as much is because other groupmates take over and have a stronger, more assertive personality; which makes it difficult for all members to feel as if they can contribute (Siew & Basari, 2024). Also conflict can arise because there are students that are placed into groups that do not work well together and this can become challenging for students to collaborate with each other (Liebech-Lien and Sjølie, 2021; Blatchford et al., 2003).

From a study done by Mertz et al. (2023), students discussed what should occur if another member was not working productively within the project. The students stated that there should be a face to face conversation as a group and if that does not induce change, then asking for consultation from the teacher (Mertz et al., 2023).

Summary

The idea of collaboration should be seen as an educational outcome that is explicitly taught and not only embedded within the content (Liebech-Lien and Sjølie, 2021). Collaborative work is when there is a common goal for the group and no single person can solve the problem independently, but through collaborating with others allows for success (Care et. al, 2016). “When teachers choose to implement CL, they need to determine how the class will be organized (i.e. composition and size of groups), the type of task (i.e., level of complexity), the mode of instruction (i.e., direct teaching or small group interaction), the patterns of communication (i.e., language needed to mediate learning), and the types of academic and social behaviors expected from the students (i.e., standards of performance and the specific interpersonal and small-group

skills required)” (Gillies & Boyle, 2010, p. 938). In order to help students establish and practice their collaborative skills, teachers should be working with students to establish collaboration expectations, assign roles, and provide multiple opportunities to practice collaboration skills (Bevo et al., 2025). Even though collaborative learning can be more enjoyable and create more motivation within learning for the students, it does not automatically lead to a better understanding of science concepts (Long et al., 2021). The teacher will need to analyze what pedagogy will be effective; teacher directed and/or student centered (Long et al., 2021). Teacher directed or student centered practices and pedagogy has potential to allow for student achievement, what determines if there is achievement tends to be the frequency of activities, but also the amount of guidance that is provided with the activities (Long et al., 2021).

Some positives of collaborative learning include: the development of content knowledge, the development of skills required to work with others, increase in motivation of own learning, and students becoming active learners (Bada & Jita, 2022). Collaborative learning is required for PBL and “learning responsibility, independence, and discipline are three outcomes of PBL” (Bell, 2010, p. 40). The more collaborative work that students participate in within the classroom, they tend to learn the skills that are needed to collaborate and will ultimately need less scaffolding (Bada & Jita, 2022; Lange et. al., 2016). When students are authentically and effectively collaborating with each other, it allows them to focus on sharing ideas and ultimately learning from each other (Bielik et al., 2022). If students are given conflict resolution strategies, a group that is struggling a teacher would need to make “a judgment call to either meet more

frequently with the group or to let the group try to work things out, depending on the severity of issues identified” (Mertz et. al., 2023, p.502).

The struggle that some teachers face is moving from the theory of collaboration to actually implementing it within the classroom because they tend to lack the experience on teaching the required collaborative skills, the amount of time required to create authentic collaborative learning projects/activities, and resources needed (Gratton, 2019; Bielik et al., 2022; Liebech-Lien and Sjølie, 2021). When teachers are implementing collaborative learning within their classrooms they struggle with effectively implementing it from theory to practice (Adl-Amini et al., 2024). According to a study done by Adl-Amini et al. (2024) in Germany discovered that teachers that struggled in implementing cooperative learning within the classroom tend to “place more emphasis on academic goals and see the training of and reflection on social skills as too time-consuming” (Adl-Amini et al., 2024, p. 2277)

Having students reflect on their collaboration skills; including how well they communicated, listened to others' ideas, and respected others' opinions, will allow the students to continually improve those skills. These types of collaboration skills are critical to being able to be successful in the global workplace. (Bell, 2010; Deepa & Seth, 2013). As students become more accountable for their own learning, they start to rely less on the teacher and the teacher role shifts to a facilitator (Bada & Jita, 2022). It is important that each group has norms of collaboration, students are reminded of behavior expectations, and students are assigned roles (Chapman & Roberts, 2015). Assigning students roles or having students create roles within collaborative learning

“create opportunities and norms for collaboration within the groups” (Chapman & Roberts, 2015, p.29).

Conclusions

Many studies have determined that students need to be explicitly taught how to work cooperatively together and how to implement collaborative projects/activities by scaffolding them (Gillies and Boyle, 2010; Bielik et al., 2022; Buchs et al., 2016).

Collaborative learning allows for a better understanding of the content if the skill is scaffolded correctly (Lange et. al., 2016). A study done by Blatchford et al. (2003) in the U.K., determined that there needs to be more and/or improved student training in group work skills.

As the classroom shifts to collaborative learning the teacher role shifts to more of a facilitator role and students should become more accountable for their own learning, following group norms, handling conflict, and being an active listener (Gillies and Boyle, 2010; Bada and Jita, 2022). The quality of collaborative learning should be modeled and practiced (Lu et al., 2021). Along with collaborative learning, the development of productive peer talk should be scaffolded, practiced, and it was recommended to use sentence starters pending the level of the students. (Hu & Chen, 2024). The study done by Adl-Amini et al. (2024), recommends that there is more training on how to “avoid a spiral of negative beliefs and unsuccessful implementation of CL” (p. 2278). Ultimately, the more collaborative learning that students partake in, the “students develop the skill of learning how to learn” (Bada & Jita, p.10, 2022). The author will be addressing potential ways that a secondary teacher could explicitly teach collaborative learning skills through the scaffolding role play skits, creation of group norms, creation of a group

contract, the development of group roles, conflict resolution strategies, and the development of a collaboration rubric that is filled out by the individual and by the group peers. This paper will not address intentional grouping of students or the development of the project/activity, but will focus on the preparation of student collaboration skills and the shift in the teacher role.

Chapter 3: Project Description

Introduction

This project focuses on the problem that students lack effective collaboration skills. Collaboration skills are considered a 21st century skill that is needed for academic and professional success. Students have to develop a foundational knowledge of the content within classrooms, they also need to know how to work within a group and effectively collaborate in order to learn from each other and provide effective feedback. It should not be assumed that students automatically have collaboration skills; this needs to be modeled, scaffolded, and explicitly taught for students (Liebech-Lien and Sjølie, 2021). Students need to know how to work cooperatively together and how to implement collaborative projects/activities (Gillies and Boyle, 2010; Bielik et al., 2022; Buchs et al., 2016).

This project is a professional development. The professional development will allow teachers to be better equipped to understand what collaborative learning is, why students need collaboration skills, and how to teach collaboration skills. The professional development will provide the teachers resources on how to develop group norms, group contracts, and group roles. It will also provide conflict resolution

strategies, collaboration lesson plan, a template of a collaboration rubric and a daily collaboration reflection sheet for students. The first section of this chapter will go over the different parts of the professional development in detail and where to find the resources in the appendix. The second section will describe how the professional development will be evaluated to determine if it was successful. The third section will explain the importance of professional development and the rationale behind developing it. The last section of this chapter will explain where this professional development could be used.

Project Components:

This project is designed to be used as a professional development to equip teachers with the ability to teach collaborative learning skills to students. To assist with facilitating professional development, a presentation is provided, as seen in Appendix A. Within the presentation, it will address the following: what is collaborative learning, why do students need collaboration skills, where can this information be used, when to teach collaboration skills, who is impacted by collaboration, and how to teach collaboration skills.

A classroom environment needs to have good routines that are taught to the students, this is true to allow for an authentic culture of collaboration to occur (Liebech-Lien & Cunha de Araújo, 2020). The creation of classroom norms and group norms will help create a respectful environment (Morcom, 2022). Within Appendix B, there is an explanation of how to create group norms. It is important to include students within the creation of the norms (Morcom, 2022). Some potential questions to help guide a class discussion on developing norms and discussing the definition of respect

can be found in Appendix B. Also in Appendix B there is an example of teacher-created norms. The group norms were influenced by the research by Morcom (2022) that stated norms should have:

- “1. Participation and right to pass- given a choice to participate
(psychological safety)
2. Attentive listening- appreciate others’ perspectives
3. Mutual respect-set high expectations for behavior
4. Using lift ups/No put downs- promote a caring classroom
5. Personal best/growth mindset- promote a positive mindset to
achieve personal best” (p.1534).

Once group norms have been established, each student should be given roles and responsibilities to learn particular content (Liebech-Lien & Cunha de Araújo, 2020). This allows for students to ensure individual accountability and positive interdependence (Liebech-Lien & Cunha de Araújo, 2020; Adl-Amini et al., 2024; Bevo et al., 2025). Positive interdependence is when the group realizes that they can not succeed without each other and individual accountability is knowing that you need to complete your role to help the group (Adl-Amini et al., 2024).

Assigning students roles or having students create roles within collaborative learning “create opportunities and norms for collaboration within the groups” (Chapman & Roberts, 2015, p.29). The group roles are being used with permission from Washington University in St. Louis: Center for Teaching and Learning. As seen in

Appendix C, the recommended group roles are: manager/facilitator, speaker/presenter, reflector/strategy analyst, recorder, and other potential roles are given as well.

Within an authentic collaborative learning project, students should have an individual aspect of the project and contribute with the group to achieve a similar goal (Bada & Jita, 2022). A group contract is an important aspect within collaborative learning because it allows students to hold each other accountable. When students are in groups, they are developing how to communicate and be successful together. This can result in conflict. Types of conflicts that might arise is someone not contributing and is just relying on the rest of the group or if someone is overtaking the group and/or groupmates take over and have a stronger, more assertive personality; which makes it difficult for all members to feel as if they can contribute (Bell, 2010; Lange et. al., 2016; Siew & Basari, 2024; Gillies & Boyle, 2010). The contract can be used to refer back to during conflict resolution and when holding students accountable with their group roles. An example of a group contract can be seen Appendix D .

As mentioned in the group contract, conflict tends to arise (Bell, 2010; Lange et. al., 2016; Siew & Basari, 2024; Gillies & Boyle, 2010). Being able to work in teams, while completing a common goal and resolving conflict is an essential to collaborative learning (Deepa and Seth, 2013). Teachers need to equip students on how to manage conflict without teacher involvement and to supply strategies that allow students to consider the perspectives of others (Gillies & Boyle, 2010). Three conflict resolution strategies that are explained within Appendix E include: rock, paper, and scissors, break, and perspectives.

Students need to be taught how to collaborate with each other and this can be done by scaffolding the skills of collaboration (Bell, 2010; Bielik et al., 2022; Gillies and Boyle, 2010; Lange et. al., 2016; Blatchford et al., 2003). The lesson found in Appendix F should be used to introduce collaboration prior to doing a collaborative learning project. However, it is important to review group norms, group contracts, group roles, collaboration rubrics, conflict resolution, and daily reflection prior to every collaborative learning project. The development of collaboration skills takes a lot of time and practice for the student and the teacher (Bevo et al., 2025)

Within the lesson (Appendix F) students will reflect on their past experiences of collaborative learning, attempt a collaborative project without any scaffolding of their skills, and discuss the importance that norms and roles have on collaborative learning. Then students will be given a collaborative learning conflict project (Appendix G). Within the project, the group will need to develop a solution to a common conflict that occurs, then present their scenario and solution, and then present on reflecting their collaborative learning ability. The common conflict scenarios can be seen in Appendix H.

A collaboration rubric is used to address the quality of content work, commitment to the group, level of leadership, communication, and amount of individual work that was done to help the success of the group; it can be used as a teaching tool on what good collaboration looks and sounds like (Mertz et al., 2023). The rubric should be scaffolded to fit the needs of the students; the rubric is used for a student to evaluate themselves and their peers (Mertz et al., 2023). An example of a collaboration rubric that could be used at secondary education level can be found in Appendix I. Along with

peers evaluating each other, it is important to have students reflect on their own skills (Lu et al., 2021). This is the reason of a daily self-reflection form that can be found in Appendix J

The rationale behind having students fill out a collaboration rubric and a daily self-reflection is that it focuses on their collaborative skills versus the content. Students can hold each other accountable and motivate each other through self and peer reflection of a rubric and evaluations done by peers (Bada & Jita, 2022; Bell, 2010; Mertz et al., 2023; Lu et al., 2021).

Project Evaluation:

The purpose of this project is to address the problem that students lack effective collaboration skills by equipping teachers with knowledge on how to teach and model those skills. The rationale behind this project is because teachers tend to lack the training needed to effectively participate in a classroom environment that is centered around collaboration (Blatchford et al., 2003). The project will be evaluated by giving a pre-survey and a post-survey. The survey can be seen in Appendix K. The same survey will be given to the participants before and after the professional development. The project will be considered successful if teachers are able to take the resources and apply them to their students.

Project Conclusions:

Collaboration skills are needed in order for students to develop skills that are needed for the 21st century (Roshid and Haider, 2024). Twenty-first century skills can be classified as critical thinking, problem solving, decision making, and collaboration (Care et.al, 2016). It should not be assumed that students know how to effectively

collaborate with peers (Liebech-Lien and Sjølie, 2021). The students will need to know the procedures on how to collaborate (Liebech-Lien & Cunha de Araújo, 2020; Naggar and Daher, 2023; Bell, 2010; Bielik et al., 2022; Gillies and Boyle, 2010; Lange et al., 2016; Blatchford et al., 2003). Developing collaboration skills increase with a curriculum that is focused on PBL and/or CL (Siew & Basari, 2024).

The students need to be provided with collaborative learning opportunities within a safe respectful environment to practice collaborative procedures (Lu et al., 2021). The environment should allow students to feel comfortable working with peers (Lange et al., 2016). If group norms are in place, there becomes an increased potential of a respectful classroom environment and effective collaborative learning (Puntambekar et al, 2023; Morcom, 2022). The professional development provides resources that allow teachers to become more equipped on how to teach those procedures and norms; it does not focus on the development of content curriculum.

When students are collaborating with peers, conflict and motivational factors can be impacted and they need to develop positive interdependence and individual accountability (Adl-Amini et al., 2024; Bevo et al., 2025). Evaluation from peers impacts a student's extrinsic motivation during collaborative learning and has potential to positively impact the students' participation (Lu et al., 2021). The professional development provides a group contract, group norms, peer-review rubric, and a self-reflection that allows for students to monitor their own motivation and hold each other accountable. Those tools should fit the needs of the students; and can be used for a student to evaluate themselves and their peers (Mertz et al., 2023).

For when conflict occurs during collaboration, groups need to be able to resolve the conflict in order to accomplish the common goal (Deepa and Seth, 2013). Teachers need to equip students on how to manage conflict without teacher involvement and to supply strategies that allow students to consider the perspectives of others (Gillies & Boyle, 2010). This professional development offers conflict resolution strategies; this can be found in appendix E.

Collaboration skills are needed when there are two or more students working together to solve a problem or to better understand the content; this is considered active student centered learning (Bada & Jita, 2022; Liebech-Lien & Cunha de Araújo, 2020). This project is a professional development that allows teachers to become better equipped to explicitly teach collaboration skills; since many research studies have found that teachers lack the training needed to structure a classroom that is centered around collaboration (Blatchford et al., 2003; Liebech-Lien and Sjølie, 2021)

Plans for Implementation

This professional development can be used during a conference for secondary education science; such as Science on the Grand in Grand Rapids, MI. The rationale behind focusing on science curriculum is due to the amount of collaborative work that takes place to accomplish Next Generation Science Standards. Overall, professional development is focusing on collaboration skills, so any area of study that has project based learning within their curriculum. The professional development should take about 60 minutes to 90 minutes. The teachers that are participating within the session will take on the role of the student during the collaboration lesson (Appendix F) and then will be applying their professional expertise during the rest of the session.

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Appendices

Appendix A

Professional Develop Presentation Slides



Learning Outcomes:

By the end of the this pd, you will be able to:

- Understand what collaborative learning is.
- Understand why students need collaboration skills.
- Explicitly teach collaboration skills using provided resources.

To teach collaboration skills, we will examine:

- The development of group norms, group contracts, and group roles
- Conflict resolution strategies
- Collaboration lesson plan
- A template of a collaboration rubric
- Daily collaboration reflection

Agenda:

- ☐ What is collaborative learning and collaboration skills?
 - ☐ Why do students need collaboration skills?
 - ☐ Where can this PD information be used?
 - ☐ When to teach collaboration skills?
 - ☐ Who is impacted by collaboration?
 - ☐ How to teach collaboration skills?

What is collaborative learning?

- 2 or more students that are working together to solve a problem in order to develop a better understanding of the content.
- Active learning curriculum
- Student centered classroom
- Students to work on complex real life projects/problems that require teamwork, communication, and problem solving; all while focusing on the content standards required by the course
- Students should have an individual aspect of the project and contribute with the group to create the bigger picture.
 - Common goal for the group and no single person can solve the problem independently, but through collaborating with others allows for success



Why collaborative learning?

- Engaging and active learning
- Higher retention rates
- More authentic learning
- Developing 21st century skills:
 - Critical thinking
 - Problem solving
 - Decision making
 - Collaboration
- Potential for students to become more intrinsically motivated about their learning
- Potential for students to seek deeper questions
- Opportunity for peers to provide effective feedback and peer mentoring

Sources: Bielik et al., 2022; Care et.al, 2016; Gratton, 2019; Naggar and Daher, 2023; Bell, 2010; Roshid and Haider, 2024

What are
collaboration
skills?



What are collaboration skills?

Collaboration skills can include:

- Communication
- Listening to others' ideas
- Respecting others' perspectives
- Adaptability
- Problem solving
- Conflict Resolution
- Time management

In order to help students establish and practice their collaborative skills, teachers should be working with students to establish collaboration expectations, assign roles, and provide multiple opportunities to practice collaboration skills (Bevo et al., 2025).

Why do we care
about
collaboration
skills?



Why do students need collaboration skills?

“In the future, children must enter a workforce in which they will be judged on their performance. They will be evaluated not only on their outcomes, but also on their collaborative, negotiating, planning, and organizing skills”
(Bell, 2010, p. 43).

- Develop skills that allow for lifelong learning and the ability to see worldwide perspectives
- A safe environment to develop 21st century skills that will be needed for professional and personal relationships.

Where can this PD information be used?

- Focusing on middle school (6-8)
 - Can be adapted for any grade level
- The teacher will need to analyze what pedagogy will be effective; teacher directed and/or student centered
- Teacher directed or student centered practices and pedagogy has potential to allow for student achievement, what determines if there is achievement tends to be the frequency of activities, but also the amount of guidance that is provided with the activities.

When to teach collaboration skills?

- Students need to know the procedures of how to collaborate before the project.
- The classroom environment needs to be established to allow students to feel comfortable working with peers that have a variety of perspectives
- Do not assume that students automatically know how to collaborate efficiently and resolve conflict with each other.
- Students need to be taught how to collaborate with each other and this can be done by scaffolding, modeling and explicitly teaching the skills of collaboration

Sources: Liebech-Lien & Cunha de Araújo, 2020; Naggar and Daher, 2023; Lange et al., 2016; Liebech-Lien and Sjølie, 2021; Bell, 2010; Bielik et al., 2022; Gillies and Boyle, 2010; Blatchford et al., 2003



How does a
teacher's role
shift during
collaborative
learning?



Who is impacted by collaboration?

- Teachers need to be concerned with students learning the content and their ability to work with others.
- As students become more accountable for their own learning, following group norms, handling conflict, and being an active listener, the teacher role shifts to a facilitator
- The teacher needs to develop more of a facilitator role and will need to know when to step in.
 - Knowing the amount of guidance that is required requires scaffolding collaboration instructions.
 - Students can develop negative feelings towards cooperative learning if the teacher is unaware of how to teach and use the pedagogy accurately.

Sources: Naidoo & Mabaso, 2023; Bada & Jita, 2022; Gillies and Boyle, 2010; Najjare and Daher, 2023; Bell, 2010; Blatchford et al., 2003; Gratton, 2019; Siew and Basari, 2024

How to teach collaboration skills?

- It should not be assumed that students know how to collaborate and it should not be assumed that teachers have been provided resources on how to teach collaboration.
- This part of the professional development will go through 6 different resources that can be adapted to help explicitly teach and scaffold collaboration skills pending on the students' abilities.

Group Norms

What:

The group norms should include a variation of the following:

- Active participation from all group members
 - Respecting all opinions and viewpoints
- High expectations of each member in terms of effort and behavior
- Not putting anybody down: promoting the respectful classroom
 - All members should be doing their best collaborating.

(Morcom, 2022).

Rationale:

The creation of classroom norms and group norms will help create a respectful environment (Morcom, 2022).

In order for a classroom to foster a culture of collaboration, students need to be taught good routines that allow for collaboration to occur
(Liebech-Lien & Cunha de Araújo, 2020)

Appendix B

Creation of Group Norms

First:

- Have a class discussion about the definition of respect and what it looks like within the classroom.
 - a. Including:
 - i. How should the teacher be treated?
 - ii. How should the students be treated?
 - iii. How to treat materials and the classroom.
 - iv. What to do when a student is not being respectful?
 - v. How to hold each other accountable?
 - vi. Why do we need one group norms?

Second:

- Either create group norms as a class that all groups have to abide by, allow groups to make their own norms, or a mix of both. The norms should be visible and revisited often. An example attached of a teacher made group norms. This could be used for all groups if students are not familiar with norms and/or collaborating.
- The group norms should include a variation of the following:
 1. Active participation from all group members
 2. Respecting all opinions and viewpoints
 3. High expectations of each member in terms of effort and behavior
 4. Not putting anybody down: promoting the respectful classroom
 5. All members should be doing their best collaborating.

GROUP NORMS

AS YOU ARE
COLLABORATING,
FOLLOW THESE
NORMS!

PARTICIPATE



Equal voices,
one voice at a
time

RESPECT



Respect all new
ideas & different
opinions

ATTENDANCE



Be on time, be
checked in, be
prepared

HIGH EXPECTATIONS



Meaningful
peer talk &
discussion

NO-PUT DOWNS



Assume positive
intentions,
be nice

DO YOUR BEST



Contribute to the
workload, apply
your knowledge, do
NOT give up

BEHAVIOR EXPECTATIONS



Follow
classroom
expectations

Group Roles

What:

Liebech-Lien and Cunha de Araujo (2020), each student was given roles and responsibilities to learn particular content

Rationale:

Allows for students to ensure individual accountability and interdependence (Liebech-Lien & Cunha de Araújo, 2020)

Assigning students roles or having students create roles within collaborative learning “create opportunities and norms for collaboration within the groups” (Chapman & Roberts, 2015, p.29).

Appendix C

The following are role cards for collaboration.

The group roles are being used with permission from Washington University in St. Louis' Center for Teaching and Learning.
Reference:
Groupwork: Overview: Center for Teaching and Learning (2022, August 5). <https://ctl.wustl.edu/resources/group-work-overview/>

Strategies when implementing group roles:

- The teacher should walk students through each role.
- Each role card explains the description of the job and includes potential sentence starters.
- The teacher can decide to allow students to pick roles, assign roles, and how often to have students switch roles.
- If a student(s) is not fulfilling their role, make sure they are being held accountable.
 - Consult the group contract, if they have one
 - Pending the situation, talk with the group as a whole or with the student individually
- Remind students of the group norms
- Scaffold the responsibilities of the roles
 - If students have never been assigned roles, make sure the students understand their roles' responsibilities by reviewing and modeling (see Collaboration Skits)
 - Once students have been using the roles often, review the expectations of each role.
- Rotate the roles frequently
- As a teacher, you are the facilitator and should be an active listener
 - Avoid interrupting and let students develop their own group dynamics
- Allow for reflection time

Appendix C

MANAGER or FACILITATOR

- Make sure team starts quickly and remains focused during the activity.
 - Assign tasks for collecting and distributing materials as needed.
 - Assign roles like calculator or significant figure checker.
 - "I think we have everything, are we ready to begin?"
- Takes care of time management.
 - Keep an eye on the clock.
 - Keep team moving forward and communicate about discussion deadlines.
 - "I think we need to focus on _____ so we complete this section on time."
 - "We have _____ minutes before we need to discuss this. Let's get this done."
- Make sure all voices in the team are heard.
 - Address team members by name and ensure that everyone contributes.
 - Assign different members to read sections of activity on a rotating basis.
 - "(Name), what do you think about . . . ?"
 - "I would like to hear what you think, (name)."

SPEAKER or PRESENTER

- Communicates team questions and clarifications with the teacher or other teams.
(This is the only team member designated to do so.)
 - "Our team is confused about how _____ relates to _____."
 - "Our team reached consensus that the answer to number _____ was _____."
- Ensures all team members have had a chance to respond before asking outside sources.
 - "Does anyone in our team know the answer to _____?"
 - "Before we ask the teacher, could someone clarify the answer to...."
- Ensures that everyone in the team agrees on what to ask if an outside source is needed.
 - "Does everyone agree we need to find out . . . ?"
- Presents conclusions of the team to the class, as requested.
 - "The reasoning we used to answer number _____ was . . ."

Appendix C

REFLECTOR or STRATEGY ANALYST

- Guides consensus-building process; team must agree on responses to questions.
 - "Would you all agree that _____ is a good answer for question number _____?"
 - "Could you please rephrase what you just said?"
 - "Is your answer completely supported by your explanation?"
 - "Would that response make sense to someone from another team?"
- Observes team dynamics and behavior with respect to the learning process.
 - Is everyone in the team participating?
 - Are team members listening carefully to each other?
 - Are team members being patient and respectful of each other?
- Reports to the team periodically during the activity on how the team performs.
 - "Let's stop for a minute. I have a couple comments on what we are doing well and a suggestion on how we could be more productive."
 - "Let's wait for (name) to catch up before we move on."
- Be ready to report to the entire class about how well the team is operating.

RECORDER

- Records the names and roles of the group members at the beginning of each activity.
- Records the important aspects of group discussions, observations, insights, etc.
 - "This seems like an important conclusion. Let's stop for a minute so I can get this into our report."
 - "That was a great insight. Do you mind (name) if I quote you in our group's report?"
- The recorder's report is a log of the important concepts that the group has learned.

Appendix C

Other potential roles:

- Encourager:
 - Asks peers to continue their thoughts
 - Asks questions that allow for deeper thinking and open discussion
- Questioner:
 - Makes sure the group is examining multiple viewpoints and not coming to a solution quickly without examining all possibilities.
- Checker:
 - Double checks group members work
 - Asks for clarification if any material does not seem to fit.

Group Contract

What:

A group contract is an important aspect within collaborative learning because it allows students to hold each other accountable.

Rationale:

Within an authentic collaborative learning project, students should have an individual aspect of the project and contribute with the group to create the bigger picture (Bada & Jita, 2022). It is important that students are held accountable for their work and peers can hold each other accountable (Lu et al., 2021).

Types of conflicts that might arise:

- A peer(s) not contributing and is just relying on the rest of the group
 - A peer(s) is overtaking the group
- A peer(s) that take over and have a stronger, more assertive personality; which makes it difficult for all members to feel as if they can contribute (Bell, 2010; Lange et. al., 2016; Siew & Basari, 2024; Gillies & Boyle, 2010)

Appendix D

Group Contract for _____ Project

Due Date _____

The success of the group will depend on the collaboration of its members. Employers will expect you to know how to collaborate within a group. This includes determining what needs to be done, how to find information, how to share the work, and how to resolve conflicts.

Collaboration includes, but not limited to:

- Following the group norms
 - Participating fully
- Being respectful to others opinions and work
 - Having an open mind
- Meeting your responsibilities

Directions: Each member will initial each item and sign the document at the end. Return it to the teacher who will make copies for your group. If you disagree with these rules, then as a group amend them. However, the teacher has the right to deny

If you can not abide by this contract, you can become "fired" from the group. If you become "fired" from the group, you will need to restart the project individually without taking any resources that your group collaborated on.

Before you are fired, you will get:

1. Verbal Warning (documented)
2. 2nd warning & teacher conference
3. Fired

Name	Role(s)

1. Each member will do their best to be at school and will meet in person or virtually on agreed upon time. **Initials:**
2. Group members who are absent, will need to talk with the group and come to agreement about how they will make up the work that they missed. **Initials:**
3. A group member who is absent more than _____times will be dismissed from the group. **Initials:**
4. The group as a whole and/or individually will not plagiarize or use AI on their project. **Initials:**
5. Each group member will complete their assigned role to the best of their ability. In the event, that a member is not able to complete their assigned role, they will [write a consequence]:

Initials:

6. Members will follow the group norms, including respect and not interrupting. This includes no name-calling, saying an idea is stupid, etc. In the event that a peer is being disrespectful, they will [write a consequence]:

Initials:

7. Members should stay by their group, should be using a noise level of a 1, and all other behavior expectations put in place. In the event that the peer can not, they will [write a consequence]:

Initials:

8. Other rules that the group would like to add:

Each group member prints their name and then signs and dates showing that they will uphold the group contract for this project. Failure to upload the contract, will result in the group member being "fired" from the group. This will require teacher intervention and ultimately becomes the teacher's choice.

Name	Signature	Date

Conflict Resolution Strategies

What:

Strategies and sentence starters for students to use to help resolve minor conflict without teacher involvement.

Rationale:

- Teachers need to equip students on how to manage conflict without teacher involvement and to supply strategies that allow students to consider the perspectives of others (Gillies & Boyle, 2010).
- Students need to develop positive interdependence and individual accountability (Adl-Amini et al., 2024; Bevo et al., 2025).
- Being able to work in teams, while completing a common goal and resolving conflict is an essential to collaborative learning (Deepa and Seth, 2013)

Appendix E



Strategy 1: Rock, Paper, Scissors

When: To settle a dispute or make an unbiased group decision. It should be decided as a group that the winner will get their choice.

How: Each participate in the game states what they want, the group should be okay with all the choices laid out. If group members have a disagreement about the choices, go to strategy 3. Play rock, paper, scissors and the winner of the game gets their choice.

Potential Sentence Starters:

- "To fairly decide who gets to [do something], let's play Rock Paper Scissors."
- "Let's make this decision with a simple round of Rock Paper Scissors."

Appendix E

Strategy 2: Break

When: If group members are becoming emotional and no decisions can be made without group members taking things personal and not assuming positive intentions.

How: Take a break on the particular part of the project and start working on a different part of the project. Once emotions have calmed or a mediator is available, return back to that part of the project.

Potential Sentence Starters:

- "Let's take a break from [certain part of the project] and come back to it when everyone has cooled down. We can work on [a different part of the project] instead."
- "I am concerned we are all getting frustrated with each other. Let's take a break from [part of the project] and work on [part of the project]."

Appendix E

Strategy 3: Perspectives

When: To make a decision when multiple group members are at a disagreement and feel passionately that they are correct and not being heard. To approach a member that is not withholding up their contract with the group.

How:

1. Take a pause
2. Each person states their perspective using “I feel [emotion] when [situation] because [reason] and I would [what you want].” This can be stated or written down.
3. Each person is able to state their perspective without cross talking or interruptions.
4. If possible, group members propose their solution either verbally or written.
5. The group votes on the best solutions or request teacher mediation.

Potential Sentence Starters:

- “We are disagreeing and we should hear everyone’s perspective.”
- “I think we are struggling to work as a team and have everyone participate within their roles. I think we should listen to each other’s perspectives about how collaborating is going.”

Collaboration Lesson

What:

Students need to be taught how to collaborate with each other and this can be done by scaffolding the skills of collaboration (Bell, 2010; Bielik et al., 2022; Gillies and Boyle, 2010; Lange et. al., 2016; Blatchford et al., 2003).

Rationale:

- The development of collaboration skills takes a lot of time and practice for the student and the teacher (Bevo et al., 2025)
 - Teachers and students tend to lack the training needed to effectively participate in a classroom environment that is centered around collaboration (Blatchford et al., 2003).
- The development of productive peer talk should be scaffolded, practiced, and there should be self and peer reflection (Hu & Chen, 2024).

See Appendix F, G, & H

Collaboration Rubric

What:

Collaboration rubric that is used to address the quality of content work, commitment to the group, level of leadership, communication, and amount of individual work that was done to help the success of the group and can be used to evaluate themselves and their peers (Mertz et al., 2023). The rubric can be used as a teaching tool on what good collaboration looks and sounds like (Mertz et al., 2023).

Rationale:

- Evaluation from groupmates impacts a student's extrinsic motivation and can hold the student accountable for their participation (Lu et al., 2021).
- Students can hold each other accountable through self and peer reflection of a rubric by peers (Bada & Jita, 2022; Bell, 2010; Mertz et al., 2023).
- "Accountability to peers often has greater consequences and provides more motivation for students than if they were only responsible to the teacher" (Bell, 2010, p. 40).

See Appendix I

Daily Self-Reflection

What:

As a daily check out and/or after the project, have the students reflect on their collaborative skills and conflict resolution skills

Rationale:

- Students need to develop positive interdependence and individual accountability (Adl-Amini et al., 2024; Bevo et al., 2025).
- The development of collaboration skills takes a lot of time and practice for the student Bevo et al., 2025)

Appendix I

Name: _____

Date: _____ Hour: _____

Self-Reflection for _____ Project

1. My role today was:

Circle the role that you had within your group today.

Manager/ Speaker/ Reflector/Strategy Recorder Other: _____
Facilitator Presenter Analyst

2. List what you and/or your group accomplished today.

3. How were you able to fulfill your role within your group?

4. Did your group have any conflicts? If so, how did you resolve them? If not, explain what went well.

5. What can you do better next time you collaborate? *Do not say "nothing." If things went great, then explain why things went well.*

Questions/Comments

Please fill out the post-survey

Appendix B

Group Norms

Creation of Group Norms

First:

- Have a class discussion about the definition of respect and what it looks like within the classroom.
 - a. Including:
 - i. How should the teacher be treated?
 - ii. How should the students be treated?
 - iii. How to treat materials and the classroom.
 - iv. What to do when a student is not being respectful?
 - v. How to hold each other accountable?
 - vi. Why do we need one group norms?

Second:

- Either create group norms as a class that all groups have to abide by, allow groups to make their own norms, or a mix of both. The norms should be visible and revisited often. An example attached of a teacher made group norms. This could be used for all groups if students are not familiar with norms and/or collaborating.
- The group norms should include a variation of the following:
 1. Active participation from all group members
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 3. High expectations of each member in terms of effort and behavior
 4. Not putting anybody down: promoting the respectful classroom
 5. All members should be doing their best collaborating.

GROUP NORMS

AS YOU ARE
COLLABORATING,
FOLLOW THESE
NORMS!

PARTICIPATE



Equal voices,
one voice at a
time

RESPECT



Respect all new
ideas & different
opinions

ATTENDANCE



Be on time, be
checked in, be
prepared

HIGH EXPECTATIONS



Meaningful
peer talk &
discussion

NO-PUT DOWNS



Assume positive
intentions,
be nice

DO YOUR BEST



Contribute to the
workload, apply
your knowledge, do
NOT give up

BEHAVIOR EXPECTATIONS



Follow
classroom
expectations

Appendix C

Group Role

The following are role cards for collaboration.

The group roles are being used with permission from Washington University in St. Louis: Center for Teaching and Learning.

Reference:

Group work: Overview. Center for Teaching and Learning. (2022, August 1). <https://ctl.wustl.edu/resources/group-work-overview/>

Strategies when implementing group roles:

- The teacher should walk students through each role.
- Each role card explains the description of the job and includes potential sentence starters.
- The teacher can decide to allow students to pick roles, assign roles, and how often to have students switch roles.
- If a student(s) is not fulfilling their role, make sure they are being held accountable.
 - Consult the group contract, if they have one
 - Pending the situation, talk with the group as a whole or with the student individually
- Remind students of the group norms
- Scaffold the responsibilities of the roles
 - If students have never been assigned roles, make sure the students understand their roles' responsibilities by reviewing and modeling (see Collaboration Skits)
 - Once students have been using the roles often, review the expectations of each role.
- Rotate the roles frequently
- As a teacher, you are the facilitator and should be an active listener
 - Avoid interrupting and let students develop their own group dynamics
- Allow for reflection time

MANAGER or FACILITATOR

- Make sure team starts quickly and remains focused during the activity.
 - Assign tasks for collecting and distributing materials as needed.
 - Assign roles like calculator or significant figure checker.
 - "I think we have everything, are we ready to begin?"
- Takes care of time management.
 - Keep an eye on the clock.
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- Make sure all voices in the team are heard.
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 - Assign different members to read sections of activity on a rotating basis.
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- Communicates team questions and clarifications with the teacher or other teams.
(This is the only team member designated to do so.)
 - "Our team is confused about how _____ relates to _____."
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- Ensures all team members have had a chance to respond before asking outside sources.
 - "Does anyone in our team know the answer to _____?"
 - "Before we ask the teacher, could someone clarify the answer to...."
- Ensures that everyone in the team agrees on what to ask if an outside source is needed.
 - "Does everyone agree we need to find out . . . ?"
- Presents conclusions of the team to the class, as requested.
 - "The reasoning we used to answer number _____ was . . ."

REFLECTOR or STRATEGY ANALYST

- Guides consensus-building process; team must agree on responses to questions.
 - "Would you all agree that _____ is a good answer for question number _____?"
 - "Could you please rephrase what you just said?"
 - "Is your answer completely supported by your explanation?"
 - "Would that response make sense to someone from another team?"
- Observes team dynamics and behavior with respect to the learning process.
 - Is everyone in the team participating?
 - Are team members listening carefully to each other?
 - Are team members being patient and respectful of each other?
- Reports to the team periodically during the activity on how the team performs.
 - "Let's stop for a minute. I have a couple comments on what we are doing well and a suggestion on how we could be more productive."
 - "Let's wait for (name) to catch up before we move on."
- Be ready to report to the entire class about how well the team is operating.

RECORDER

- Records the names and roles of the group members at the beginning of each activity.
- Records the important aspects of group discussions, observations, insights, etc.
 - “This seems like an important conclusion. Let’s stop for a minute so I can get this into our report.”
 - “That was a great insight. Do you mind (name) if I quote you in our group’s report?”
- The recorder’s report is a log of the important concepts that the group has learned.

Other potential roles:

- Encourager:
 - Asks peers to continue their thoughts
 - Asks questions that allow for deeper thinking and open discussion
- Questioner:
 - Makes sure the group is examining multiple viewpoints and not coming to a solution quickly without examining all possibilities.
- Checker:
 - Double checks group members work
 - Asks for clarification if any material does not seem to fit.

Appendix D

Group Contract

Within an authentic collaborative learning project, students should have an individual aspect of the project and contribute with the group to create the bigger picture (Bada & Jita, 2022). A group contract is an important aspect within collaborative learning because it allows students to hold each other accountable. When students are in groups, they are developing how to communicate and be successful together. This can result in conflict. Types of conflicts that might arise is someone not contributing and is just relying on the rest of the group or if someone is overtaking the group and/or groupmates take over and have a stronger, more assertive personality; which makes it difficult for all members to feel as if they can contribute (Bell, 2010; Lange et. al., 2016; Siew & Basari, 2024; Gillies & Boyle, 2010). The contract can be used to refer back to during conflict resolution and when holding students accountable with their group roles.

The following is one example of a group contract. The contract can be written by the teacher only, by group members or both. It can be the teacher's decision if all groups have the same contract, let groups design their own, or groups can construct their own but have a set of expectations that all groups have to follow. However it is decided to implement the group contract, it needs to be explicitly taught to the students, time for reflection, and scaffolded. The parts where the group will need to write in a consequence will need to be guided.

Group Contract for _____ Project

Due Date _____

The success of the group will depend on the collaboration of its members. Employers will expect you to know how to collaborate within a group. This includes determining what needs to be done, how to find information, how to share the work, and how to resolve conflicts.

Collaboration includes, but not limited to:

- Following the group norms
 - Participating fully
- Being respectful to others opinions and work
 - Having an open mind
- Meeting your responsibilities

Directions: Each member will initial each item and sign the document at the end. Return it to the teacher who will make copies for your group. If you disagree with these rules, then as a group amend them. However, the teacher has the right to deny

If you can not abide by this contract, you can become “fired” from the group. If you become “fired” from the group, you will need to restart the project individually without taking any resources that your group collaborated on.

Before you are fired, you will get:

1. Verbal Warning (documented)
2. 2nd warning & teacher conference
3. Fired

Name	Role(s)

1. Each member will do their best to be at school and will meet in person or virtually on agreed upon time. **Initials:**
2. Group members who are absent, will need to talk with the group and come to agreement about how they will make up the work that they missed. **Initials:**
3. A group member who is absent more than _____times will be dismissed from the group.**Initials:**
4. The group as a whole and/or individually will not plagiarize or use AI on their project. **Initials:**
5. Each group member will complete their assigned role to the best of their ability. In the event, that a member is not able to complete their assigned role, they will [write a consequence]:

Initials:

6. Members will follow the group norms, including respect and not interrupting. This includes no name-calling, saying an idea is stupid, etc. In the event that a peer is being disrespectful, they will [write a consequence]:

Initials:

7. Members should stay by their group, should be using a noise level of a 1, and all other behavior expectations put in place. In the event that the peer can not, they will [write a consequence]:

Initials:

8. Other rules that the group would like to add:

Each group member prints their name and then signs and dates showing that they will uphold the group contract for this project. Failure to upload the contract, will result in the group member being “*fired*” from the group. This will require teacher intervention and ultimately becomes the teacher’s choice.

Name	Signature	Date

Appendix E

Conflict Resolution Strategies



Strategy 1: Rock, Paper, Scissors

When: To settle a dispute or make an unbiased group decision. It should be decided as a group that the winner will get their choice.

How: Each participate in the game states what they want, the group should be okay with all the choices laid out. If group members have a disagreement about the choices, go to strategy 3. Play rock, paper, scissors and the winner of the game gets their choice.

Potential Sentence Starters:

- "To fairly decide who gets to [do something], let's play Rock Paper Scissors."
- "Let's make this decision with a simple round of Rock Paper Scissors."

Strategy 2: Break

When: If group members are becoming emotional and no decisions can be made without group members taking things personal and not assuming positive intentions.

How: Take a break on the particular part of the project and start working on a different part of the project. Once emotions have calmed or a mediator is available, return back to that part of the project.

Potential Sentence Starters:

- "Let's take a break from [certain part of the project] and come back to it when everyone has cooled down. We can work on [a different part of the project] instead."
- "I am concerned we are all getting frustrated with each other. Let's take a break from [part of the project] and work on [part of the project]."

Strategy 3: Perspectives

When: To make a decision when multiple group members are at a disagreement and feel passionately that they are correct and not being heard. To approach a member that is not withholding up their contract with the group.

How:

1. Take a pause
2. Each person states their perspective using "I feel [emotion] when [situation] because [reason] and I would [what you want]." This can be stated or written down.
3. Each person is able to state their perspective without cross talking or interruptions.
4. If possible, group members propose their solution either verbally or written.
5. The group votes on the best solutions or request teacher mediation.

Potential Sentence Starters:

- "We are disagreeing and we should hear everyone's perspective."
- "I think we are struggling to work as a team and have everyone participate within their roles. I think we should listen to each other's perspectives about how collaborating is going."

Appendix F

Collaboration lesson

This lesson should be used to introduce collaboration prior to doing a collaborative learning project. However, it is important to review group norms, group contracts, group roles, collaboration rubrics, conflict resolution, and daily reflection prior to every collaborative learning project.

Collaboration Skills Lesson Plan

Grade Level: 6-8

Time Frame: 3 to 4 days

LEARNING OBJECTIVE:

Students will be able to develop skills needed to collaborate in groups, understand the importance of collaboration, identify roles within a team, and resolve conflicts through discussion and practice.

Success Criteria:

I can practice collaborating with my peers while upholding my role as a team.

I can attempt to respectfully resolve conflict within my group without the teacher involvement.

Key Concepts:

- **Collaboration:** Understanding collaboration as working together towards a common goal.
- **Importance of Collaboration:** Discussing the benefits of collaboration in various contexts (academic, social, and professional).
- **Conflict Resolution Strategies:** Exploring methods such as active listening, compromising, and seeking mediation.
- **Group Roles:** Understanding different roles individuals can play in a team and how these contribute to success. Roles include: manager/facilitator, speaker/presenter, reflector/strategy analyst, recorder. Other roles can be: encourager, questioner, or checker.

• Group Norms: Establishing agreements that guide group interactions and responsibilities.		
	Directions	Materials
Engage Time Frame: ½ day	1. Ask students to think back to a time when they were required to work within a group. Remind students as they are creating their lists that they should be being respectful to students and teachers and avoid using any names. a. Individually: i. Create a list of what went well. ii. Create a list of what went poorly b. With a partner: i. Instruct students to go through each list. ii. Have each student go through the 'well' list. One student should go through their whole list and then the partner goes. There should be no interrupting or cross-talking. iii. After each partner went through their list, discuss any similarities and/or differences. iv. Have the partner group do the same thing with the 'poorly' list. c. As a class: i. On chart paper, create two class lists. One list of what went 'well' and what went 'poorly' in the past. ii. Have a class discussion to reflect on their past experiences and what would have been done to make them more enjoyable. 2. Questions to ask: a. <i>"Why do you think it is important to practice being in a group?"</i>	Chart Paper Markers

	i. <i>Potential answers: Life skill, work with people outside of school</i> b. <i>“What are some things that are needed for group work to go well?”</i> i. <i>Potential answers: everyone has a role, set behavioral expectations, etc.</i> c. <i>“Before starting a group project, what should the group discuss?”</i> i. <i>Let the class discuss this answer, do not give answers. The goal of this question is for students to see the importance of norms and roles.</i> d. <i>“If you are working in a group and there is a conflict, what might you do as a group?”</i> i. <i>Have the students discuss this respectfully, but do not give out any answers.</i>	
Explore Time Frame: ½ day to 1 day	1. Build a Tower Instructions a. Place 4 students in a group. b. Each group is given 20 spaghetti noodles, 1 yard of masking tape, 1 yard of string, 1 large marshmallow , 1 pair of scissors. c. The roles of building the tower: i. The tower can not be attached to the table, it has to stand on its own. ii. The height of the tower is measured to the top of the marshmallow. The marshmallow must be whole (no missing pieces) iii. Groups do not need to use all the materials and they can use them however the group decides. d. *There should be no technology used!!* 2. Work time: a. Give students 20 minutes to work.	Per group: 20 spaghetti noodles 1 yard of masking tape 1 yard of string 1 large marshmallow 1 scissor Class: Timer Group

	3. Measurement: a. Measure each group from the base of the tower to the top of the marshmallow. 4. Group Reflection: (It might be beneficial to develop a worksheet or a form to have groups fill in with the following questions.) a. As a group, answer the following questions: i. <i>What was our strategy?</i> ii. <i>What did we do well?</i> iii. <i>What did we do poorly?</i> iv. <i>Did everyone participate?</i> v. <i>What roles did each member have?</i> vi. <i>What conflicts did we have? Did we resolve those conflicts?</i> 5. Class Discussion: a. Have each group share out with the questions.	reflection Qs.
Explain Time Frame: ½ day to 1 day	1. Define collaboration and the importance of it within academics and real world scenarios. 2. Explain what collaboration norms mean. a. See Appendix B on how to develop collaboration norms. b. Class discussion: i. <i>“Think back to building a tower, did your group follow any of the collaboration norms that were discussed?”</i> 1. <i>If not, what would have gone smoother if you followed those norms.</i> 2. <i>If yes, what norms did you follow?</i>	Per Student: Collaboration Norms Group Roles Group Contract Conflict Resolution Strategies

	3. Explain what group roles are. a. See Appendix C b. Go over the different roles with the students. c. Have students get back into their groups. Have students discuss the following questions. i. <i>“Think back to building a tower, to answer the following questions:</i> 1. <i>Did you fulfill one of those roles?</i> 2. <i>Did every group member have a role?</i> 3. <i>Would the work have gone smoother if everyone was held accountable for their role within the group?</i> d. Class discussion: i. Discuss the roles that were used during the building of a tower and what roles were not used. 4. Explain what a group contract is. a. <i>Why do we think it is important that a contract is used during collaborative learning?</i> b. See Appendix D c. Let the students know that a group contract will be revisited when you start the collaborative learning project. 5. Class discussion: a. <i>What type of conflict might arise during collaborative work?</i> b. <i>What are some ways to resolve the conflict, WITHOUT teacher involvement?</i> c. <i>Why is it important to resolve conflict that comes up</i>	
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	during collaborative work? d. <i>Why is conflict resolution an important skill to develop?</i> 6. Give each student the conflict resolution strategies. a. See Appendix E	
Extend/ Elaborate Time Frame: 1 to 2 days	1. Have students be in the same group as the “Build a Tower” 2. Give each group: a. Group role cards (Appendix C) b. Collaboration norms (Appendix B) i. <i>Or have each group create their own norms</i> c. Group contract (Appendix D) 3. Have each group work through their contract. Once a group is completed with their contract, the teacher should visit the group and have the group explain their norms, roles, and the contract. 4. Prior to giving groups their scenario, have a class discussion about the importance of contracts, norms, and roles. 5. Explain to the class that each group will be given a scenario: a. As a group, you need to develop a solution to your scenario and present it to the class. Then you will reflect on your group’s ability to collaborate. i. See Appendix G for student directions ii. See Appendix H for potential conflict scenarios 6. Before having groups present, explain to the class what an active respectful audience member looks like. 7. Pending on time, have each group present to the whole class or pick a few groups to work through their presentation.	Group: Role Cards Collaborati on norms Group Cards Conflict Scenario Optional: Poster board Chart Paper Markers White- boards
Evaluate ½ to 1 day	1. Explain the importance of peer evaluation on collaboration. Give each student a collaboration rubric (Appendix I) a. Read each row with the students and ask the	Per Student: Collaborati

	students what that looks like and means b. Individually: i. Have students fill out a column for each of your group members and for yourself. Provide the rationale behind your rating. Make sure you are being RESPECTFUL when giving your rationale. Your teacher will go through the rubric and will return feedback provided by your peers on how you could improve. This is confidential and can impact your individual grade. 2. Class Discussion: a. <i>Why do you think it is important to have a rubric that focuses on collaboration and not on the content that we are learning?</i> b. <i>Do you think you will be great at collaborating right away?</i> c. <i>How do you think you could improve on collaborating with your peers?</i> d. <i>Why is it so important to practice collaborative learning?</i> e. <i>Do you only use collaborative learning within an academic setting?</i> 3. Have students complete the daily reflection on collaboration. Students should reflect on building a tower and skits/presentations when completing the reflection sheet. a. See Appendix J	on rubric Daily Reflection
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Appendix G

Collaborative Learning Conflict Project

Collaborative Learning Conflict Project

Student Instructions

Directions:

As a group, you need to develop a solution to your scenario and present it to the class. Then you will reflect on your group's ability to collaborate.

1. Scenario presentation:

- ☐ Description of the scenario
- ☐ How the conflict was resolved
 - **This should be demonstrated through a short skit****
- ☐ What collaboration norms were broken and/or within the conflict and resolution?

2. Ability to collaborate reflection presentation:

- ☐ How did each member fulfill their group role?
- ☐ What norms were most important to your group?
- ☐ What, if any, aspects of the contract did you have to refer back to?
- ☐ As a group, did you have any conflicts?

Ways to present:

- You can use Google Slides, Canva, Chart Paper, Whiteboards, etc.

Time Frame:

- Work time: 30-60 minutes
- Presentation time: 2-4 minutes

Grading:

- Individual Grade:
 - Your group members and teacher will be assessing your ability to collaborate work.
- Collaborative Learning Grade:
 - Your teacher will be looking for the 7 aspects of the presentation and the group will be assessed.

Appendix H

Conflict Scenarios

Conflict Scenarios

Collaborative Learning Conflict Project

Directions:
As a group, you need to develop a solution to your scenario and present it to the class. Then you will reflect on your group's ability to collaborate.

- Scenario presentation:**
 - Description of the scenario
 - How the conflict was resolved
 - *This should be demonstrated through a short skit*
 - What collaboration norms were broken within the scenario
- Ability to collaborate reflection presentation:**
 - How did each member fulfill their group role?
 - What norms were most important to your group?
 - What, if any, aspects of the conflict did you have to refer back to?
 - As a group, did you have any conflicts?

Ways to present:

- You can use Google Slides, Canvas, Chart Paper, Whiteboards, etc.

Time Frame:

- Work time: 30-60 minutes
- Presentation time: 2-4 minutes

Grading:

- Individual Grade:
 - Your group members and teacher will be assessing your ability to collaborate work.
- Collaborative Learning Grade:
 - Your teacher will be looking for the 7 aspects of the presentation and the group will be assessed.

Conflict 1:

Your family just moved into a new house. There are three rooms available for you, your brother and sister, but one is larger than the others and has a bigger closet. Your sister has the most clothes and insists she needs the room. Your brother thinks he should get the room because he's the oldest. You want the extra space for your drum set. It bothered everyone when you practiced in the dining room. Your parents told you to work it out amongst yourselves.

3

Conflict 2:

The school is engaging students in an anti-drug campaign. You and Dana have been chosen to create a large banner to be hung in the school's main hallway. Dana wants to draw a series of student portraits, each with their own drug awareness slogan. You don't like drawing and would rather use the banner to explain the school's campaign in large block letters.

4

Conflict 3:

You are assigned to a group with social studies. You are not looking forward to working in this group because there are two students, Jacob and Alex, that are constantly disrupting and talking over the teacher. When you and Becky try to start the project, Jacob and Alex just keep talking off task and disrupting other groups. The teacher has warned that the group needs to stay focused and is not following behavior expectations.

5

Conflict 4:

In Language Arts, everybody had to do the same reading individually and then come together as a group to develop a model about the reading. Bethann is in your group and she did not do the reading and is not contributing with all the other members and is relying on all the other members for completion of the project. Bethann asked you if she could copy your individual aspect of the project.

6

Conflict 5:

You are working on a presentation within your group. The work was split up between the four of you and everyone has their part of the project to complete. However, TJ keeps messing up with everyone's work and saying "It's just a joke." The other group members are getting frustrated that their hard work is being destroyed and that they have to redo their work.

7**Conflict 6:**

Jennifer is in your group for an Art project. Jennifer is a nice student, however, she likes to do all the work and does not like to listen to others' opinions. When you try to give an idea about how to complete the work, she says "no" and continues to do the project without the help or ideas of any other member of the group.

8**Conflict 7:**

You are in a group for Math class. All of the group members have different ideas on how to best complete a project, resulting in arguments about how to complete the problem. All group members talk over each other, interrupt each other, and can not come to an agreement about how to start the project.

9

Appendix I

Collaboration rubric

Directions: You will need to fill out a column for each of your group members and for yourself. Provide the rationale behind your rating. Make sure you are being RESPECTFUL when giving your rationale. Your teacher will go through the rubric and will return feedback provided by your peers on how you could improve. This is confidential and can impact your individual grade.

Criteria	3 - Exceeds Expectations	2 - Meets Expectations	1 - Needs Improvement	Peer Name:	Peer Name:	Peer Name:	Self Evaluation:
Individual Contribution	Completes individual portion thoroughly with clear, accurate work and thoughtful input to the project.	Completes individual portion adequately with mostly accurate work and sufficient input.	Completes individual portion partially or with minimal effort; work lacks clarity or accuracy.	Rating: _____ Rationale:	Rating: _____ Rationale:	Rating: _____ Rationale:	Rating: _____ Rationale:
Team Collaboration	Actively listens, shares ideas clearly, supports peers, and helps reach consensus for the solution.	Participates in discussions, shares ideas, and cooperates with peers to complete the solution.	Rarely participates in team discussions, shows little cooperation, or hinders group progress.	Rating: _____ Rationale:	Rating: _____ Rationale:	Rating: _____ Rationale:	Rating: _____ Rationale:
Problem Solving & Solution	Contributes creative ideas and solutions; works with peers to develop a complete and effective solution.	Contributes relevant ideas and collaborates to develop a workable solution.	Offers few ideas and contributes little to solution development; solution is incomplete or ineffective.	Rating: _____ Rationale:	Rating: _____ Rationale:	Rating: _____ Rationale:	Rating: _____ Rationale:
Communication & Respect	Communicates respectfully and effectively throughout the project, encouraging others' contributions.	Communicates respectfully and listens to others, maintaining a positive team environment.	Communication is often unclear, disrespectful, or disrupts team harmony.	Rating: _____ Rationale:	Rating: _____ Rationale:	Rating: _____ Rationale:	Rating: _____ Rationale:
Responsibility and Time Management	Consistently meets deadlines and encourages team to stay on	Meets deadlines and usually stays focused during group work.	Frequently misses deadlines or distracts the team from	Rating: _____ Rationale:	Rating: _____ Rationale:	Rating: _____ Rationale:	Rating: _____ Rationale:

Appendix J

Daily Self-Reflection

Name: _____

Date: _____ Hour: _____

Self-Reflection for _____ Project

1. My role today was:

Circle the role that you had within your group today.

Manager/ Speaker/ Reflector/Strategy Recorder Other: _____
Facilitator Presenter Analyst

2. List what you and/or your group accomplished today.

3. How were you able to fulfill your role within your group?

4. Did your group have any conflicts? If so, how did you resolve them? If not, explain what went well.

5. What can you do better next time you collaborate? *Do not say "nothing." If things went great, then explain why things went well.*

Appendix K

Pre & Post Evaluation Survey

Pre and Post Collaboration Skill Survey

By the end of this PD, you will be able to:

- Understand what collaborative learning is.
- Understand why students need collaboration skills.
- Explicitly teach collaboration skills using provided resources.

To teach collaboration skills, we will examine:

- The development of group norms, group contracts, and group roles
- Conflict resolution strategies
- Collaboration lesson plan
- A template of a collaboration rubric
- Daily collaboration reflection

I have a clear understanding of what collaborative learning entails.

5 - Strongly Agree 4 - Agree 3 - Neutral 2 - Disagree 1 - Strongly Disagree

I recognize the importance of collaboration skills for student success.

5 - Strongly Agree 4 - Agree 3 - Neutral 2 - Disagree 1 - Strongly Disagree

I feel confident in teaching collaboration skills to my students.

5 - Strongly Agree 4 - Agree 3 - Neutral 2 - Disagree 1 - Strongly Disagree

The resources provided for teaching collaboration skills are useful.

5 - Strongly Agree 4 - Agree 3 - Neutral 2 - Disagree 1 - Strongly Disagree

I understand how to develop group norms effectively.

5 - Strongly Agree 4 - Agree 3 - Neutral 2 - Disagree 1 - Strongly Disagree

I am familiar with strategies for conflict resolution in group settings.

5 - Strongly Agree 4 - Agree 3 - Neutral 2 - Disagree 1 - Strongly Disagree

I feel prepared to implement a collaboration lesson plan in my classroom.

5 - Strongly Agree 4 - Agree 3 - Neutral 2 - Disagree 1 - Strongly Disagree

I find the collaboration rubric template to be helpful.

5 - Strongly Agree 4 - Agree 3 - Neutral 2 - Disagree 1 - Strongly Disagree

I believe daily reflections can enhance students' collaboration skills.

5 - Strongly Agree 4 - Agree 3 - Neutral 2 - Disagree 1 - Strongly Disagree

I am eager to apply what I learned about collaboration skills in my teaching practice.

5 - Strongly Agree 4 - Agree 3 - Neutral 2 - Disagree 1 - Strongly Disagree

Appendix L

Permissions

Dear Whom it May Concerned,

I am currently enrolled in the Grand Valley State University (GVSU), Advanced Studies in Education Program, and I am writing a Master's Project for the completion of my Master's Degree in Education. My project is entitled "Collaboration Skills Professional Development" May I receive permission to include in the appendixes of my Master's Project a copy(ies) of the following item(s)?

Using Roles in Group Work. I would mainly like to use the role cards and place cards as example of group roles that can be used to help students effectively collaborate with each other. (<https://ctl.wustl.edu/resources/using-roles-in-group-work/>)

Your signature at the bottom portion of this letter confirms your ownership of the above item. The inclusion of your copyrighted material will not restrict your re-publication of the material in any other form. Please advise if you wish a specific copyright notice to be included on each page. My project may be cataloged in the GVSU library and will be available to other students and colleges for circulation.

Sincerely,

Robin VanderWel
7025 West 104th St. Fremont MI
bentlrob@mail.gvsu.edu
616-460-3252

Hi Robin: I appreciate your interest in our CTL resources. You may include them as part of your thesis project. Good luck as your research proceeds.

Eric

Eric J. Fournier, Ph.D.

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Pronouns: he/him/his

PERMISSION IS GRANTED to you Robin VanderWel to include the requested material(s) in their GVSU Master's of Education Project.

Name: Robin VandeWel

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